

UNIVERSITY SCHOOL OF AUTOMATION & ROBOTICS (USAR)



**Guru Gobind Singh Indraprastha University, East Delhi
Campus, Surajmal Vihar, Delhi 110092**

**Electronic Design Automation for VLSI Lab
(ARI-354)**

**Submitted to:
Dr. Manoj Kumar
Asst. Professor, USAR**

**Submitted by:
Ronak Dutta
01419051723
IIOT - B1**

INDEX

S No.	Topic	Date	Sign
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			
9.			
10.			

Experiment 1A

AIM: Introduction to SPICE Simulator

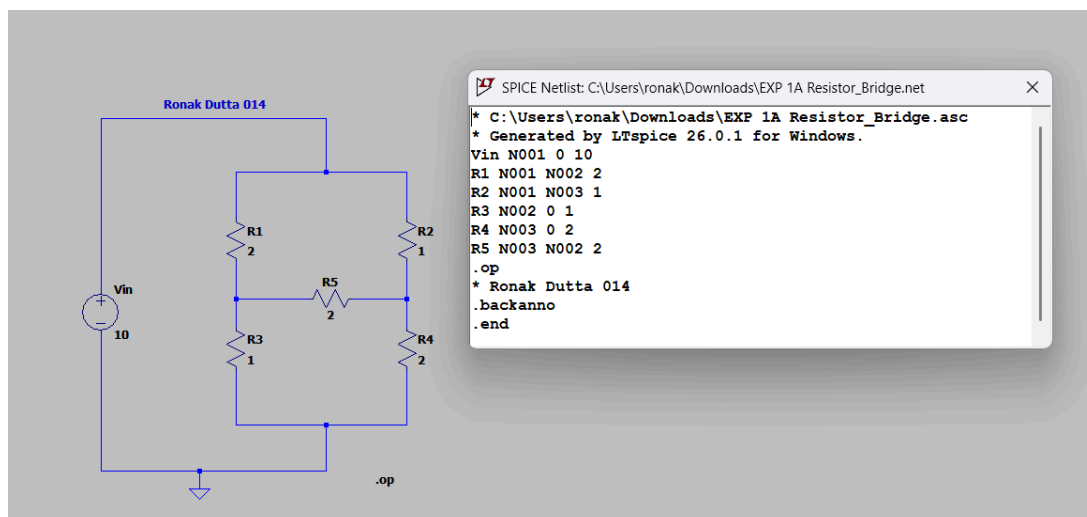
A) Register Bridge

ABOUT LT SPICE : LTspice is a free and powerful circuit simulation software developed by Analog Devices, widely used by engineers, students, and hobbyists to design, analyze, and test electronic circuits. It allows users to simulate both analog and digital circuits, such as amplifiers, power supplies, and filters, using a variety of components like resistors, capacitors, transistors, and op-amps. LTspice helps in performing different types of analysis, including DC, AC, and transient simulations, making it easier to study voltages, currents, and overall circuit behavior before physically building the design. Known for its speed, accuracy, and extensive component library, LTspice is especially popular in power electronics and analog circuit design, offering a cost-effective way to verify and optimize circuits virtually.

A) REGISTER BRIDGE

A register bridge in electronics typically refers to a circuit arrangement that uses registers (or resistors) to form a bridge configuration, commonly known as a resistive bridge or Wheatstone bridge. It is used to measure unknown resistance, detect small changes in resistance, or balance a circuit. The basic design consists of four resistors connected in a diamond shape, with a voltage source applied across one diagonal and the output (usually measured with a voltmeter or an amplifier) taken across the other diagonal.

When the bridge is balanced (meaning the ratios of the resistors are equal), the output voltage across the bridge is zero. However, if there is a small change in one resistor (due to temperature, strain, or other factors), the balance is disturbed, and a small voltage appears across the output, which can be measured and analyzed. This principle is widely used in sensors like strain gauges, temperature sensors, and pressure transducers. Register bridges are key in precision measurement applications because they can detect very tiny changes in resistance with high accuracy.



Experiment 1B

AIM: Intro to SPICE Simulator

B) Diode Rectifier

THEORY:-

What is a Diode Rectifier?

A diode rectifier is a circuit that converts alternating current (AC)—which reverses direction periodically—into direct current (DC), which flows in only one direction. At the heart of it is the diode, a device that only allows current to pass in one direction. Think of it like a one-way gate.

Half-Wave Rectifier

Working:

- AC Input has two halves: a positive half and a negative half.
- A half-wave rectifier uses only one diode.
- During the positive half cycle, the diode is forward-biased (conducts electricity), and the current flows to the load.
- During the negative half cycle, the diode is reverse-biased (blocks current), and no current flows.
- So, you only get half the wave hence the name "half-wave".

Output:

- Pulsating DC, but only for the positive half. The other half is missing.
- Full-Wave Rectifier
- Working:
- Utilizes both halves of the AC input.
- There are two types of full-wave rectifiers:
 - 1. Center-Tap Transformer with Two Diodes
 - 2. Bridge Rectifier with Four Diodes (most common)

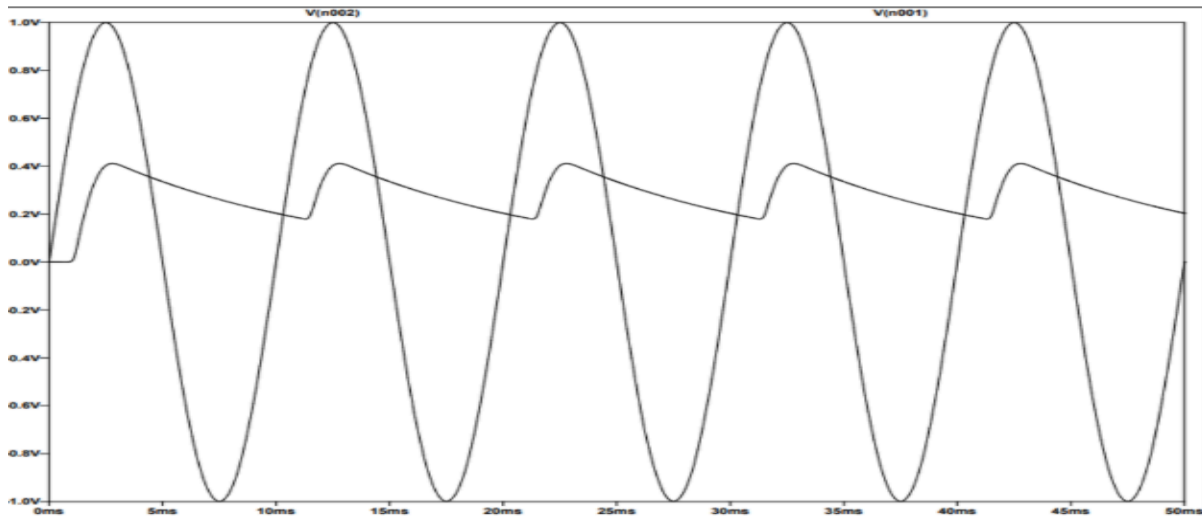
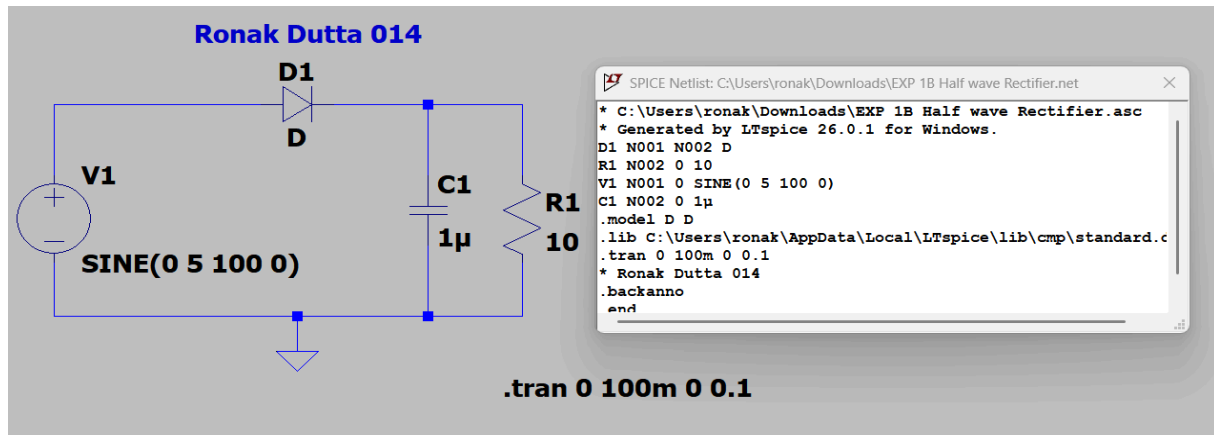
Bridge Rectifier:

- Four diodes are arranged in a bridge.
- During both halves of the AC input, current flows through the load in the same direction.
- It's smartly wired so the direction of flow is corrected every half cycle.

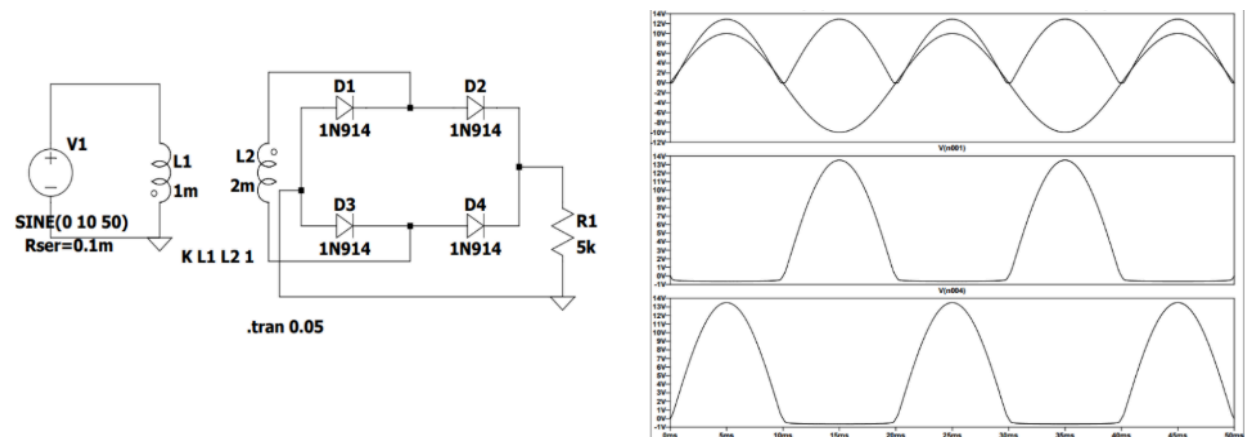
Output:

- You get a continuous pulsating DC much better than half-wave.
- Still not smooth DC, but both halves are utilized.

HALF WAVE RECTIFIER



FULL WAVE RECTIFIER



Experiment 2

AIM: Draw the DC characteristics of Nmos & Pmos Transistor

- 1) Transfer characteristics
- 2) Output characteristics

Theory :

1. NMOS Transistor Theory

- A positive voltage is applied to the Gate.
- A conductive channel forms when $V_{GS} > V_{TH}$ (i.e., gate-to-source voltage exceeds the threshold voltage).
- Electrons are the majority carriers.

2. PMOS Transistor Theory

- A negative voltage is applied to the Gate (with respect to the Source).
- A conductive channel forms when $V_{SG} > |V_{TH}|$ (i.e., source-to-gate voltage exceeds the magnitude of threshold voltage).
- Holes are the majority carriers.

1) Transfer Characteristics

- NMOS: I_D vs. V_{GS} (at constant V_{DS})
- PMOS: I_D vs. V_{SG} (at constant V_{SD})

Shows how the drain current changes with gate voltage.

2) Output Characteristics

- NMOS: I_D vs. V_{DS} (for various V_{GS})
- PMOS: I_D vs. V_{SD} (for various V_{SG})

Shows how the drain current behaves with varying drain-source voltage.

Key Formulas :

For NMOS (in Saturation):

$$I_D = (1/2) \times k_n \times (V_{GS} - V_{TH})^2$$

$$\text{Where: } k_n = \mu_n \times C_{ox} \times (W / L)$$

For PMOS (in Saturation):

$$I_D = (1/2) \times k_p \times (V_{SG} - |V_{TH}|)^2$$

$$\text{Where: } k_p = \mu_p \times C_{ox} \times (W / L)$$

Procedure:

A) For Transfer Characteristics (NMOS Example)

1. Set $V_{DS} = 5V$ (constant).
2. Vary V_{GS} from 0V to 5V in steps (e.g., 0.5V).
3. Measure I_D at each V_{GS} .
4. Plot I_D vs. V_{GS} .

B) For Output Characteristics (NMOS Example)

1. Set $V_{GS} = 2V$ (repeat for 3V, 4V).
2. Vary V_{DS} from 0V to 5V.
3. Measure I_D at each V_{DS} .
4. Plot I_D vs. V_{DS} .

Circuit Diagrams:

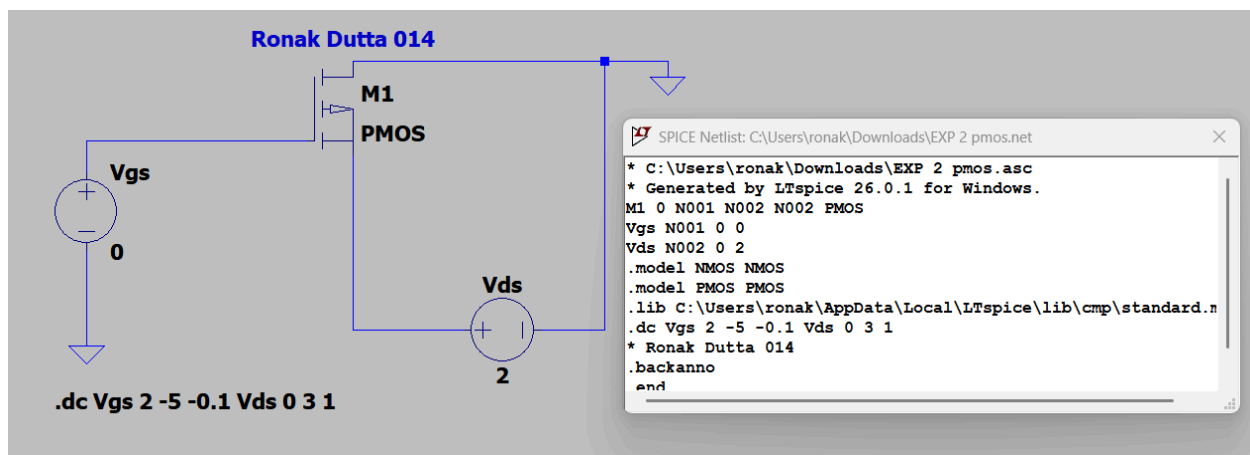
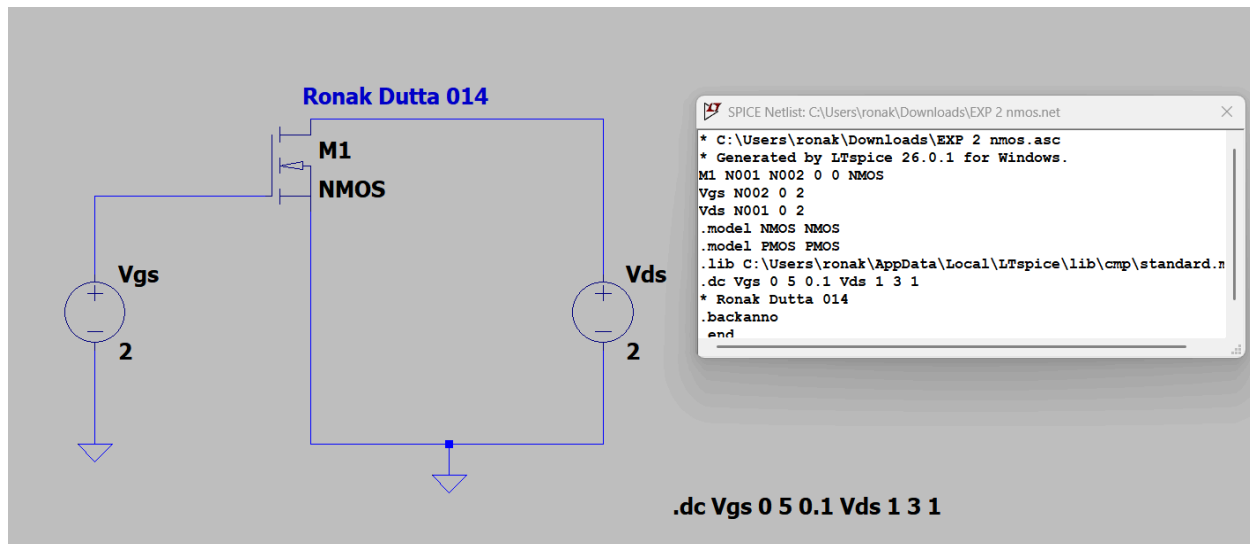
1. NMOS Test Setup:

- Connect drain to V_{DD} through a resistor.
- Source to ground.
- Gate voltage controlled via a DC source.

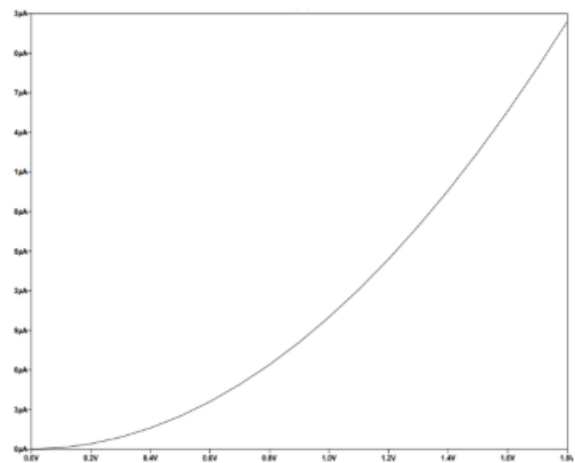
2. PMOS Test Setup:

- Source connected to V_{DD} .
- Drain connected through a resistor to ground.
- Gate voltage controlled to negative side with respect to source.
- Gate voltage controlled to negative side with respect to source.

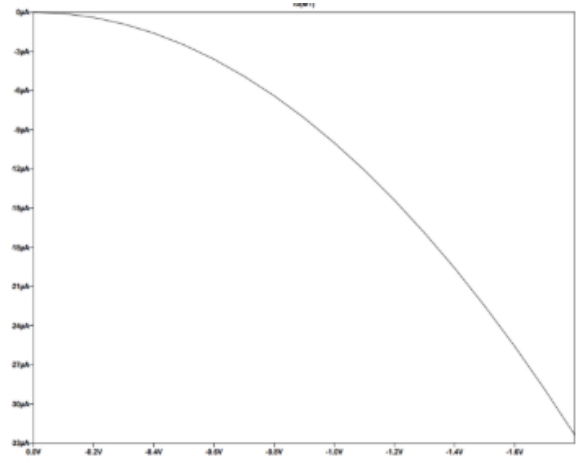
OUTPUT



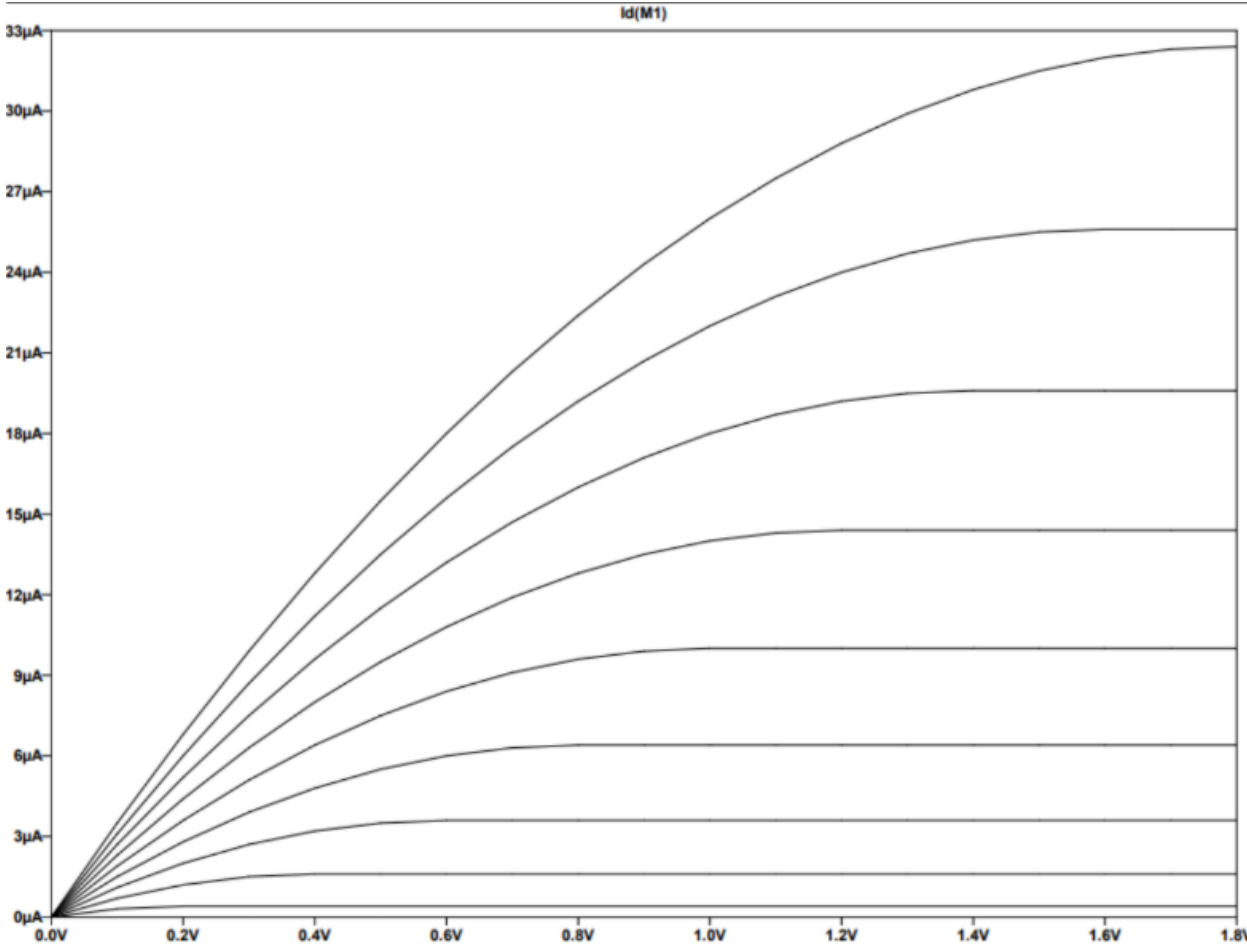
NMOS TRANSFER CHARACTERISTICS



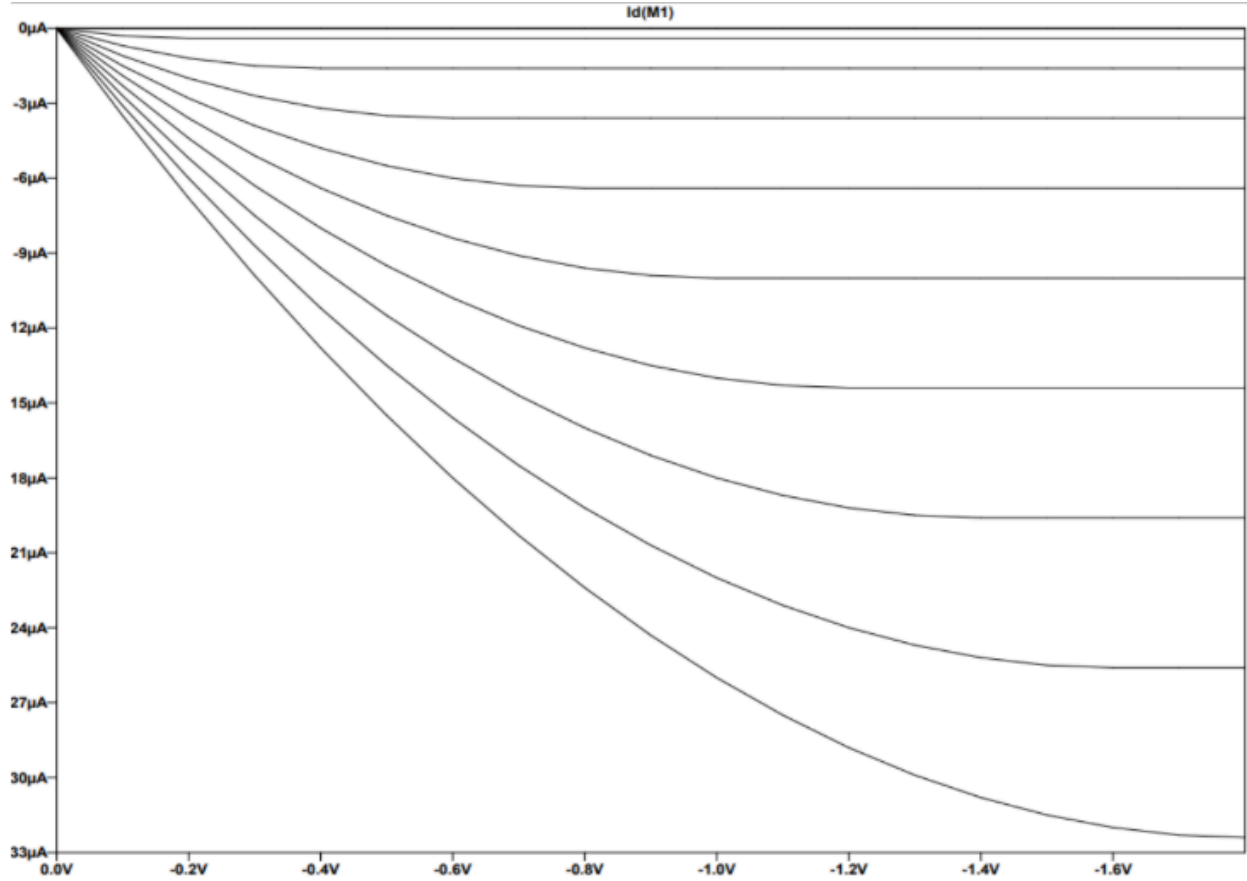
PMOS TRANSFER CHARACTERISTICS



NMOS OUTPUT CHARACTERISTICS



PMOS OUTPUT CHARACTERISTICS



Experiment 3

AIM: Draw the DC characteristics of Cmos Inverter Transistor

- 1) Transfer characteristics
- 2) Output characteristics

Theory :- What is a CMOS Inverter?

A CMOS inverter (Complementary Metal-Oxide-Semiconductor inverter) is the most fundamental logic gate used in digital electronics. It performs the logical NOT operation, outputting the opposite logic level of its input.

It consists of:

- A PMOS transistor (connected to VDD)
- An NMOS transistor (connected to GND)

Both transistors are controlled by a common gate input (V_{in}), and their drains are tied together to form the output (V_{out}).

Working Principle:

Vin Level	PMOS Status	NMOS Status	Vout
0V (Low)	ON	OFF	VDD (High)
VDD (High)	OFF	ON	0V (Low)
Mid Voltage	Partially ON	Partially ON	Depends

Transistor Operation Regions:

As V_{in} increases from 0 to VDD, the transistors move through various operating regions:

1. Region 1 ($V_{in} = 0$ to V_{tn}):
 - NMOS is OFF (cutoff)
 - PMOS is ON (saturation)
 - Output = VDD (logic high)
2. Region 2 ($V_{in} = V_{tn}$ to $VDD - |V_{tp}|$):
 - NMOS enters saturation, PMOS enters saturation or linear
 - Both transistors conduct
 - Output starts dropping
 - This is the transition region, where switching happens
3. Region 3 ($V_{in} \geq VDD - |V_{tp}|$):
 - NMOS is ON (linear)
 - PMOS is OFF (cutoff)
 - Output = 0V (logic low)

Transfer Characteristics (Vout vs Vin):

The transfer characteristic curve of a CMOS inverter is:

- High at low input voltage
- Sharp transition around $V_{in} = V_{DD}/2$
- Low at high input voltage

This curve gives insight into:

- Noise margins
- Threshold voltage (V_{th})
- Switching behavior

V_{th} is the input voltage at which $V_{out} = V_{in}$. This is where the inverter starts changing state.

Power Considerations:

- CMOS inverters consume nearly zero static power because:
 - Only one transistor conducts at a time (except during switching).
- Power is only consumed during transition (dynamic power) due to charging and discharging of load capacitance.

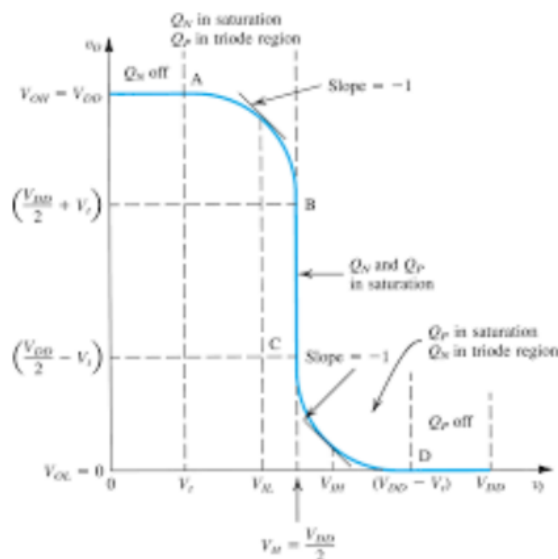
Output Characteristics (Iout vs Vout):

The output characteristics of a CMOS inverter show how much current is delivered at different output voltages.

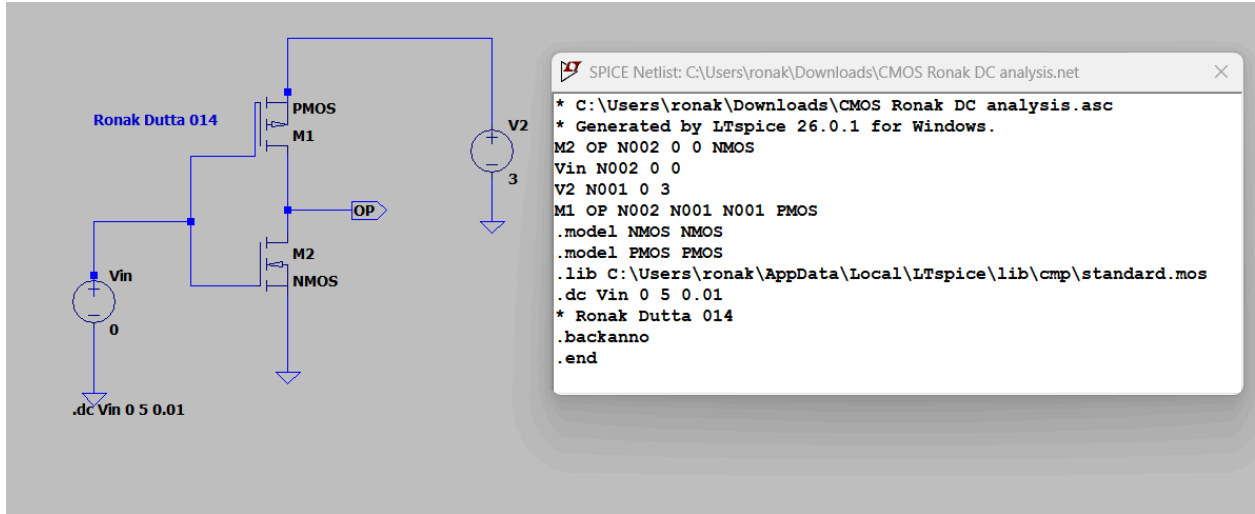
- When V_{in} is constant, and V_{out} is varied, the current drawn by the load changes.
- This tells us the driving capability of the inverter, important for connecting to multiple inputs (fan-out).

Advantages of CMOS Inverter:

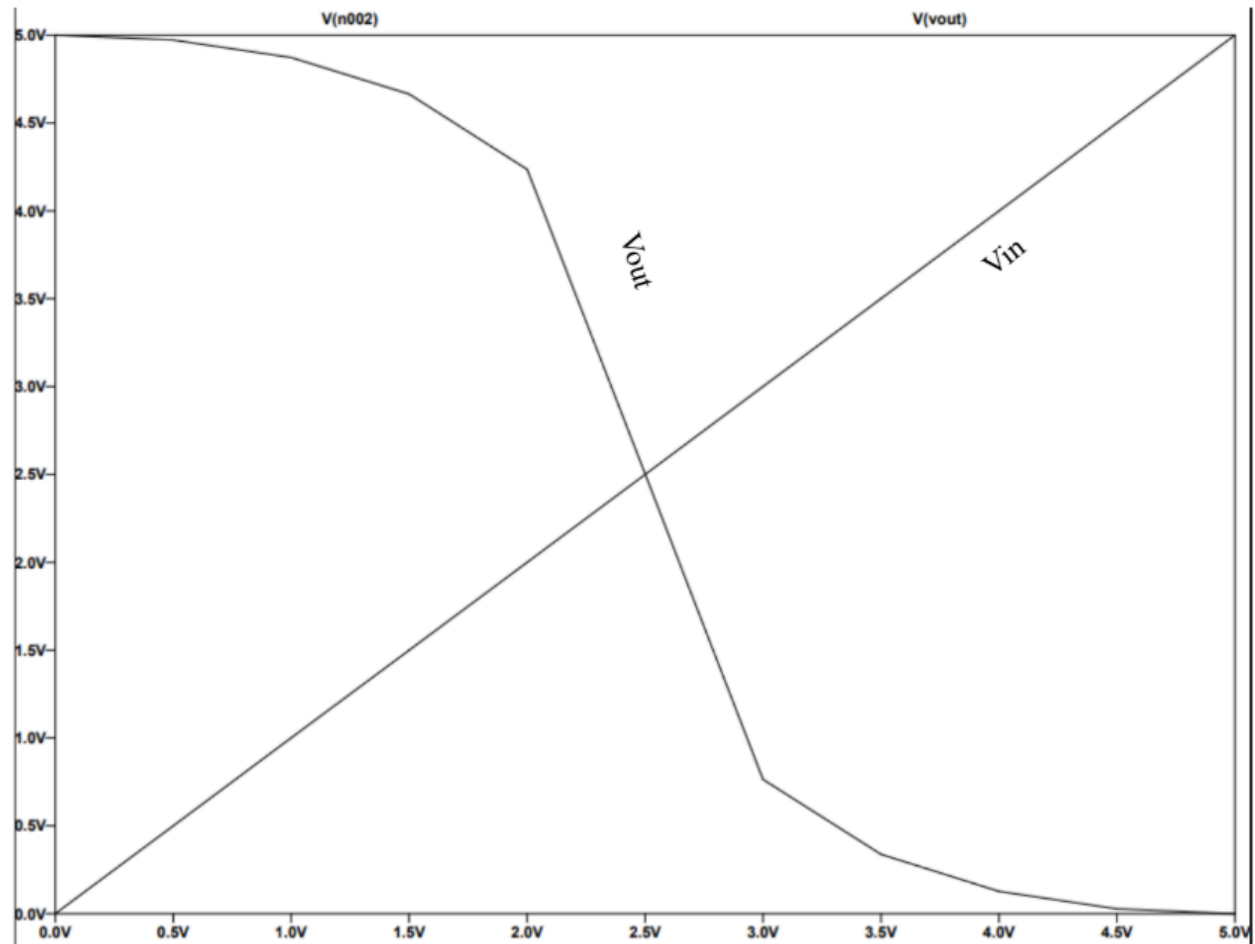
- High noise immunity
- Low static power dissipation
- Fast switching
- High input impedance
- Symmetric output characteristics



CIRCUIT DIAGRAM:



OUTPUT



Experiment 4

AIM: Design and simulate a CMOS

- 1) NAND Gate**
- 2) NOR Gate**

Theory :

CMOS Logic and Gates

CMOS (Complementary Metal-Oxide-Semiconductor) logic combines PMOS and NMOS transistors to build logic gates with:

- Low power consumption
- High noise immunity
- Balanced pull-up (PMOS) and pull-down (NMOS) structures

CMOS NAND Gate

- Logic: $Y = (AB)'$
- Structure:
 - PMOS: Parallel (Vdd to output)
 - NMOS: Series (output to GND)
- Operation: Output is LOW only when both A and B are HIGH; otherwise, HIGH

CMOS NOR Gate

- Logic: $Y = (A + B)'$
- Structure:
 - PMOS: Series (Vdd to output)
 - NMOS: Parallel (output to GND)
- Operation: Output is HIGH only when both A and B are LOW; otherwise, LOW

CMOS NAND Gate

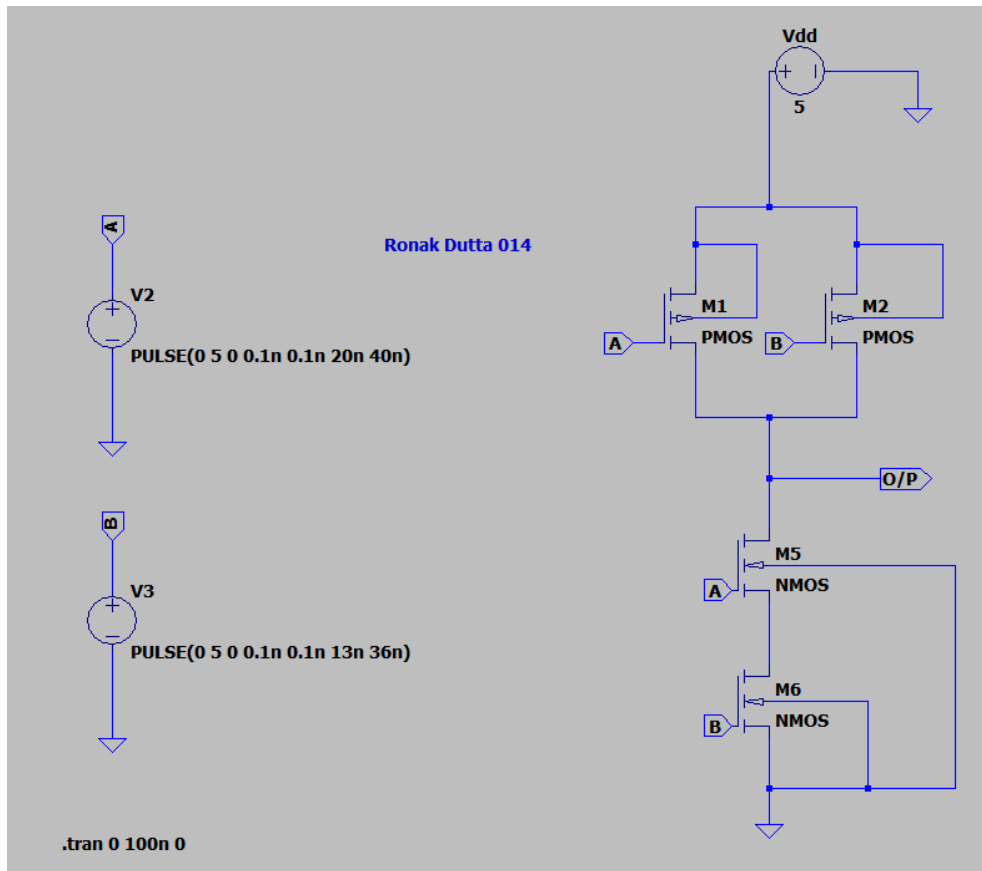
A	B	Output ($Y = (AB)'$)
0	0	1
0	1	1
1	0	1
1	1	0

CMOS NOR Gate

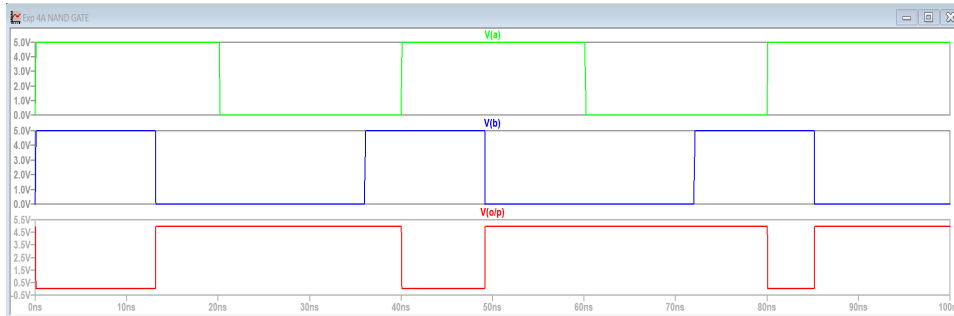
A	B	Output ($Y = (A + B)'$)
0	0	1
0	1	0
1	0	0
1	1	0

a) NAND GATE

CIRCUIT DIAGRAM:



OUTPUT:

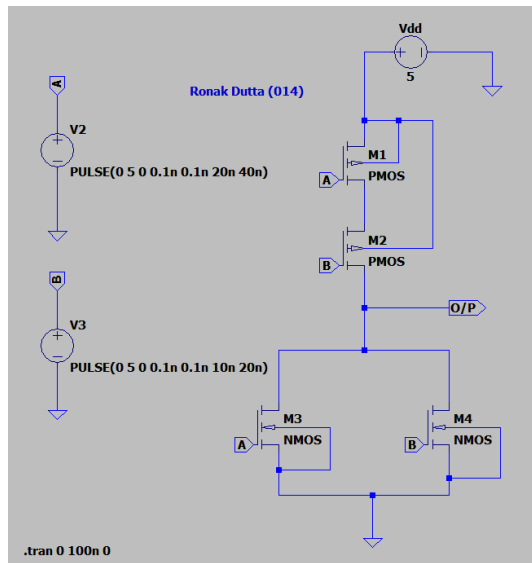


NETLIST:

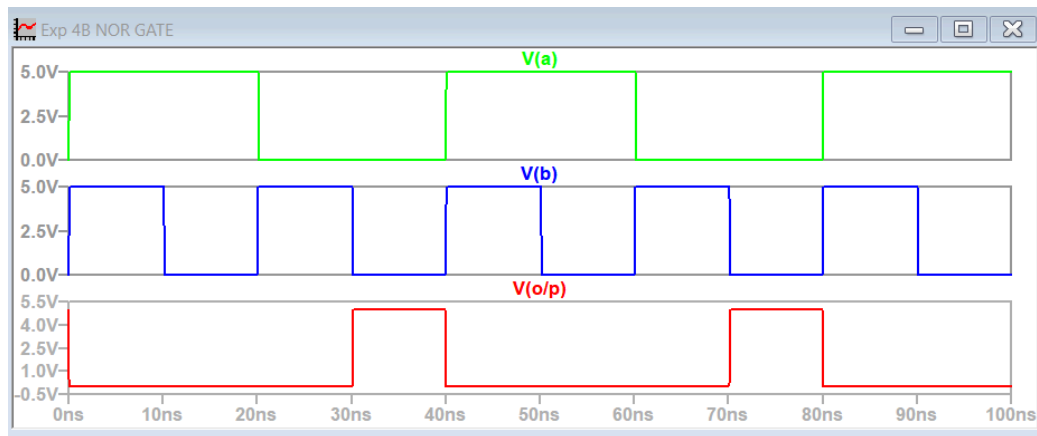
```
SPICE Netlist: C:\Users\ronak\Downloads\Exp 4A NAND GATE.net
* C:\Users\ronak\Downloads\Exp 4A NAND GATE.asc
* Generated by LTspice 26.0.1 for Windows.
V2 A 0 PULSE(0 5 0 0.1n 0.1n 20n 40n)
V3 B 0 PULSE(0 5 0 0.1n 0.1n 13n 36n)
Vdd N001 0 5
M5 O/P A N002 0 NMOS
M6 N002 B 0 0 NMOS
M1 N001 A O/P N001 PMOS
M2 N001 B O/P N001 PMOS
.model NMOS NMOS
.model PMOS PMOS
.lib C:\Users\ronak\AppData\Local\LTspice\lib\cmp\standard.mos
.tran 0 100n 0
* Ronak Dutta 014
.backanno
.end
```

b) NOR Gate:

CIRCUIT DIAGRAM:



OUTPUT:



NETLIST:

```
SPICE Netlist: C:\Users\ronak\Downloads\Exp 4B NOR GATE.net

* C:\Users\ronak\Downloads\Exp 4B NOR GATE.asc
* Generated by LTspice 26.0.1 for Windows.
M1 N001 A N002 N001 PMOS
M2 N002 B O/P N001 PMOS
Vdd N001 0 5
M3 O/P A 0 0 NMOS
M4 O/P B 0 0 NMOS
V2 A 0 PULSE(0 5 0 0.1n 0.1n 20n 40n)
V3 B 0 PULSE(0 5 0 0.1n 0.1n 10n 20n)
.model NMOS NMOS
.model PMOS PMOS
.lib C:\Users\ronak\AppData\Local\LTspice\lib\cmp\standard.mos
.tran 0 100n 0
* Ronak Dutta (014)
.backanno
.end
```